



HeeraMon

Job Ingestion Specification

Ingestion Specification Document

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1 Purpose

This document defines how HeeraMon discovers, collects, normalizes, validates, deduplicates, and stores job opportunities from external sources.

The goal is to create a reliable internal job dataset that the matching and agent systems can use consistently.

2 Ingestion Flow

The ingestion pipeline is:

**Discover → Fetch → Extract → Normalize → Validate → Deduplicate →
Store → Evaluate**

Job ingestion should be separated from job matching. The ingestion system produces clean, structured job records; HeeraMon then evaluates those records against user preferences.

3 Job Sources

HeeraMon should support a source-adaptor approach so new sources can be added without changing the rest of the system.

Potential source types include:

- Public job APIs
- RSS/Atom feeds
- Public job boards
- Company career pages
- Structured job postings / JSON-LD
- Other legally and technically accessible sources

The initial implementation should begin with a small number of reliable and accessible sources rather than attempting to cover the entire internet.

Each source adapter should have a consistent interface for retrieving job data.

4 Job Normalization

Different sources may describe the same job differently.

HeeraMon should convert incoming postings into a common internal format containing, where available:

- Job title
- Company

- Description
- Responsibilities
- Required skills
- Preferred skills
- Salary / compensation
- Location
- Work arrangement
- Working hours
- Employment type
- Experience requirements
- Education requirements
- Benefits
- Application URL
- Source
- Original posting date
- Last-seen timestamp
- Job expiration/status information

Fields that are unavailable should remain unknown/null, not be fabricated.

5 Deduplication

The same opportunity may appear across multiple sources.

HeeraMon should detect likely duplicates using available identifiers and normalized attributes such as:

- source/job ID;
- application URL;
- company;
- normalized job title;
- location;
- posting similarity.

Duplicate detection should preserve source information where useful rather than creating multiple independent opportunities.

6 Validation and Quality

Before a job becomes available for matching, HeeraMon should perform basic validation.

At minimum:

- Job title should exist.
- Company should be identifiable where possible.
- Description should contain usable information.
- Application destination should be valid where applicable.
- Clearly malformed records should be rejected.
- Expired or unavailable jobs should be marked appropriately.

HeeraMon should preserve uncertainty instead of filling gaps with guesses.

7 Freshness

Job postings change over time.

The ingestion system should track:

- first discovered time;
- last checked time;
- source update time when available;
- active/expired status;
- changes to important fields.

Repeated ingestion should update existing records rather than blindly creating new ones.

8 Source Reliability

Sources should be treated independently.

A source may become:

- temporarily unavailable;
- rate-limited;
- structurally changed;
- stale;
- inaccessible;
- inconsistent.

A failure in one source should not stop the entire ingestion system.

Source-specific errors should be logged and observable.

9 Legal and Operational Constraints

HeeraMon should only use job information through sources and access methods that are permitted for the intended use.

The ingestion system should respect applicable:

- Terms of Service;
- robots/access restrictions;
- API limits;
- rate limits;
- copyright and data-use requirements.

The system should prefer APIs, feeds, public structured data, and permitted access methods over fragile or aggressive scraping.

10 Boundaries

The initial release does not need:

- dozens of job-source integrations;
- complete internet-wide coverage;
- sophisticated web crawling;
- real-time ingestion;
- perfect duplicate detection;
- advanced job-data enrichment;
- paid job-data infrastructure.

The architecture should make additional source adapters possible later without redesigning the matching system.

11 Success Criteria

The ingestion system is successful when HeeraMon can:

1. Collect jobs from several usable sources.
2. Convert different source formats into one internal job format.
3. Detect and reduce duplicate opportunities.
4. Preserve unknown information honestly.
5. Track job freshness and availability.
6. Handle individual source failures gracefully.
7. Provide clean job records to the matching and agent systems.

Core principle: HeeraMon does not need every job on the internet. It needs a sufficiently reliable stream of structured, usable opportunities for the personal job-search loop to work.

12 Document Control

Version	Date	Author / Role	Changes
1.0	September 30, 2026	Product / Engineering	Initial release

Approval: This document defines the job ingestion behavior for HeeraMon. All subsequent design and implementation work must remain consistent with the principles stated herein.

HeeraMon flies to help people find job that fits not only their career, but the life they want to live.



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